

Introduction to Heat transfer

Assignment

ECAM-2025_26

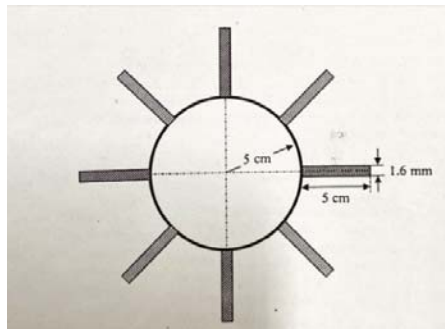
Exercise 1

Water flows on the inside of a steel pipe [$k = 43 \text{ W/m} \cdot ^\circ\text{C}$] with an ID of 2.5 cm. The wall thickness is 2 mm, and the convection coefficient on the inside is $500 \text{ W/m}^2 \cdot ^\circ\text{C}$. The convection coefficient on the outside is $12 \text{ W/m}^2 \cdot ^\circ\text{C}$. Calculate the overall heat transfer coefficient U . What is the main parameter determining U factor?

The pipe above is covered with a layer of asbestos [$k = 0.18 \text{ W/m} \cdot ^\circ\text{C}$] while still surrounded by a convection environment with $h = 12 \text{ W/m}^2 \cdot ^\circ\text{C}$. Calculate the critical insulation radius. Will the heat transfer be increased or decreased by adding an insulation thickness of (a) 0.5 mm and (b) 10 mm?

Exercise 2

A pipe with i.d. = 52 mm and o.d. = 60 mm has eight rectangular longitudinal fins welded to the outer surface. The fins are of 5 cm length and 1.6 mm thickness. The wall temperature of the pipe is 160°C . The finned tube loses heat to a flowing stream of air with a surface heat transfer coefficient of $115 \text{ W/m}^2 \cdot ^\circ\text{C}$. The thermal conductivity of the fin material is $46 \text{ W/m} \cdot ^\circ\text{C}$. Neglecting heat transfer from the edges of the fins, calculate (a) the fin efficiency, and (b) the rate of heat transfer to the air per meter length of the finned pipe. Take the temperature of the atmospheric air 25°C .



Exercise 3

A 2-cm diameter tube is maintained at constant wall temperature of 90°C . Water enters the tube at 40°C and leaves at 60°C . If the entering velocity is 3 m/s, calculate the length of tube necessary to accomplish the heating.

Hint : Take the average temperature of the water $T_{\text{water}} = (40+60)/2 = 50^\circ\text{C}$.

The fluid properties at $T_m = (90+50)/2 = 70^\circ\text{C}$ are

$$\text{Pr} = 2.54 \quad \mu = 4 \times 10^{-4} \text{ kg/m} \cdot \text{s} \quad k = 0.664 \text{ W/m} \cdot ^\circ\text{C} \quad C_p = 4.174 \text{ kJ/kg} \cdot ^\circ\text{C} \quad \rho = 978 \text{ kg/m}^3$$

Exercise 4

A fine wire having a diameter of 0.02 mm is maintained at a constant temperature of 54°C by an electric current. The wire is exposed to air at 1 atm and 0°C (free convection). Calculate the electric power necessary to maintain the wire temperature if the length is 50 cm.

Properties of air at 27°C :

$$\beta = 1/300 = 0.00333 \quad \nu = 15.69 \times 10^{-6} \text{ m}^2/\text{s} \quad k = 0.02624 \text{ W/m} \cdot ^\circ\text{C} \quad \text{Pr} = 0.708$$